Constrained Joint Decoding over Boundary-Head Candidates: Data Scaling, Replay, and Three Silent Defects

William Roe[1] (whr778@gmail.com), **Gemini**[2], and **Claude**[3] (noreply@anthropic.com)

[1] Project author and maintainer · [2] AI assistant (Google, Gemini) — diagnosis, architecture, experiments, and drafting · [3] AI assistant (Anthropic, Claude Opus 5) — design, implementation, and drafting

Paper 2 of the research programme stated in `RESEARCH_PROGRAM.md`. Revision of 2026-11-05 (results locked 2026-08-28; §5 updated to cross-reference the event-record, junction, and coreference resolution in `PAPER_3_DOCUMENT_EVENTS_AND_COREF.md`).

Companion to `PAPER_0_FOUNDATION.md` (the substrate) and `EKF_MHT_DESIGN.md` (the temporal instantiation of the same thesis: the two papers are the combinatorial and temporal scopes of one top-K inference over a shared `candidate_scores → JointProblem` contract). The build record — decision log, wiring map, increment status, cost model, deferred Phase B plan — is kept verbatim in `JOINT_IE_DESIGN_RECORD.md`. Under the programme's graduation rule this paper carries only verified, reproducible claims, including the negatives that scope them.

Abstract

A boundary-head extractor scores span, mention and pair candidates per chunk, then a **greedy per-query decode** independently thresholds each one into a final record set. We replace that decode with **constrained joint selection** over the same candidate scores and measure what it buys, on a multilingual multi-task base trained from a raw mmBERT encoder at four data scales (10K–137K records). Three results. First, on document-level relation extraction (Re-DocRED, 500-document blind test) joint selection beats greedy by **+0.052 strict F1 (+18% relative)** when both arms are read at their own swept threshold — and **beam width 1 is optimal**, monotonically beating every wider beam, so the gain comes from the *formulation* rather than from search. Second, warm-starting the 137K base on new-domain data with **30% exact replay** reverses catastrophic forgetting into a net gain, holding the original capability while nearly tripling on the new distribution. Third, and the reason we report the negative space as carefully as the positive: three separate silent defects — a decode path that dropped one schema key, a per-row split randomiser, and an uncalibrated decode threshold — each produced results that looked like clean scientific findings and were not. We report the corrected numbers, the wrong conclusions we drew before correcting them, and the standing rules adopted so they do not recur.

Contributions

1. **Constrained joint selection beats greedy per-query decoding on document-level relations** by +0.052 strict F1 (+18% relative) on the Re-DocRED 500-document blind test, with both arms read at their own swept threshold (§4).

2. **The gain is formulation, not search.** Beam width 1 is optimal and monotonically beats every wider beam, so the improvement comes from decoding under typed constraints rather than from exploring more hypotheses (§4c).
3. **30% exact replay reverses catastrophic forgetting into a net gain**, holding the original structure capability while nearly tripling performance on the new distribution — and the new corpora supplied no record-head supervision at all, so the gain is transfer through the span representations (§0b–0c).
4. **Three silent defects each produced a clean-looking finding that was wrong**: a decode path that dropped one schema key, a per-row split randomiser that scattered duplicate documents across train/validation/test, and an uncalibrated decode threshold that read a working head as exactly zero. We report the corrected numbers, the conclusions drawn before correction, and the standing rules adopted so they do not recur (§3).

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1. Thesis

The boundary head emits candidate scores per chunk — span boundaries, mention logits, pair logits, and contextual candidate states. A greedy per-query decode then selects a record set by thresholding each query independently. That is adequate for sparse, single-document extraction. The claim tested here is that it is not adequate for dense structure:

> **Global structured inference over the boundary candidate scores — not the greedy per-chunk decode — is what dense document-level extraction requires.**

Events and relations are two faces of one mechanism, which is why a single decode covers both: in the joint problem an event is a *trigger node plus role edges* and a relation is a *plain edge*. A win on only one face is a weaker but still reportable result, and the honest negative — that global decoding helps relations and not events, or the reverse — localises where greedy per-query decoding actually costs you.

The same top-K hypothesis machinery, applied *across* documents rather than within one, is the EKF/MHT tracker of the companion paper. The boundary head is the shared substrate: it removed the span architecture's 19-instances-per-type cap that had made either formulation impossible to express.

2. Method

Base models. mmBERT-base trained `from_encoder` (fresh extraction heads) on a multi-task pool of events, relations, entities, classification and structures, at nested sizes of 10K / 40K / 100K / 137K records. The 137K point is reached by adding ~37K non-leaking relation records to the 100K event pool, which warms the relation head at no generation cost. `docred` is **excluded** from the pool because Re-DocRED re-annotates the same documents and including it would leak the downstream evaluation; §3.2 verifies the exclusion held.

Decode arms. Both arms are an eval-time switch on one trained model, so no training difference can confound the comparison:

- *greedy* — each query thresholded independently, the shipped default.
- *joint* — constrained selection over the candidate scores, maximising summed edge score under typed constraints, with a beam of width W. The implementation keeps greedy as a floor, so the objective score is monotone in W by construction.

Metrics. Strict micro-F1 (exact match on type and span) unless stated. Relaxed variants and per-category supports are reported where they change the reading.

3. Experimental integrity

Three defects in this project produced results that looked clean and were not. Each is reported here with what it cost, because the failure modes generalise beyond this work.

3.1 A per-row split randomiser (found 2026-08-18)

The split writer drew a random number **per row** rather than per document, so a document emitted more than once — normal in these corpora, where one document yields several task records — had its copies scattered independently across train, validation and test. Nothing crashed; no metric looked anomalous.

A repo-wide gate found **45 corpora** shipping overlapping splits; repair dropped **21,553 records of 2.3M (0.94%)** under the precedence *test* > *val* > *train*. This configuration additionally paired regenerated training files against frozen validation slices, adding 252 train-in-val and 22 val-in-test documents.

Every number in this paper post-dates that repair. The scaling curve was re-run from scratch on repaired data; all four points now gate clean. Numbers from before the repair are not quoted here, and any earlier version of this document that quoted them is superseded.

3.2 Verifying the downstream is genuinely blind

Because the headline result is read on Re-DocRED, the `docred` exclusion was verified rather than assumed. Over the 137K training pool (129,155 distinct documents) against the Re-DocRED evaluation splits, matched on normalised document text:

split	documents	overlap with the 137K training pool
`redocred.test`	500	**0 (0.0%)**
`redocred.val`	500	**0 (0.0%)**

3.3 A decode path that silently dropped a schema key

`structure` read exactly 0.0000 on all four scaling points and on the warm start — five independent measurements, all exactly zero. The natural reading was that the record head was broken or untrainable, and this document previously said so and recommended abandoning structure data.

That reading was wrong. The inference runtime rebuilt each schema through a builder that *does* produce `record_metadata`, then copied only two other keys out of the result. With the metadata missing, the spec compiler returns an empty mapping, the record head decodes nothing, **and no error is raised**. One dropped

key produced five clean-looking zeros and one confidently wrong conclusion. §4.4 reports what the head actually does.

Two further instances of the same failure were found and are recorded because the pattern, not the instance, is the lesson: the fluent schema builder never emitted the key at all unless the caller passed a mode argument, so a caller following the obvious API also got silence; and the corpora used for the warm start emit structures without the metadata the training path needs, so records that appear to supply structure supervision supply none.

3.4 The matched-threshold rule

No arm comparison is readable until both arms are at their own swept threshold. This changed a conclusion three separate times in this experiment before being adopted as a standing rule. The largest case: at a fixed threshold of 0.5, joint decoding appeared to beat greedy by +0.106 strict F1 — but 0.5 is near greedy's *worst* operating point, and at best-vs-best the real gap is +0.052. The fixed-threshold comparison overstated the finding by a factor of two.

The corollary is that a metric read at an uncalibrated operating point is not a measurement. The record head's own decode thresholds had never been swept by any tool, and at the 0.5 default the head scores 0.0000 to 0.0760 depending on scale against 0.0238 to 0.1119 at its swept threshold.

4. Results

4.1 Joint decoding beats greedy, and the beam should be width 1

Re-DocRED, 96 relation types, one trained model, eval-time decode switch. Both arms swept over the threshold grid; each read at its own optimum. Both peak at 0.2.

threshold	greedy	joint (W=1)
0.1	0.2785	0.3270
0.2	**0.2835**	**0.3357**
0.3	0.2270	0.3264
0.4	0.1648	0.2828
0.5	0.0980	0.2406

Joint selection wins by +0.052 strict F1 (+18% relative) at best-vs-best, and beats greedy at every threshold on the grid.

The beam-width sweep is the more interesting result:

W	1	2	4	8	16	32	64
relation strict F1	**0.2406**	0.2290	0.2260	0.2211	0.2170	0.2152	0.2058

Monotonically decreasing. Widening the beam drops predictions 157 → 117, of which **18 were correct** — 45% precision on the dropped set against 61% overall — so precision rises while F1 falls. Entity metrics are byte-identical at every width, because node admission does not consult beam state; width touches only edges.

This is **score-versus-F1 divergence**. The beam maximises its objective *better* as it widens — the implementation keeps greedy as a floor, so the score is monotone — and the objective is not F1. A better search on a mis-specified objective produces worse output.

So the gain is the formulation, not the search. W=1 barely searches and wins outright. The working contrast is *independent thresholding versus constrained joint selection*, not *greedy versus beam*, and the original framing of this experiment as a beam-width study was mis-specified.

For context, OneIE (Lin et al., 2020) used θ=10 with per-step branching capped at 2, and its released package defaults to 5. Two independent global-IE decoders landed at 5–10 where our optimum is 1. The mechanism differs — their cap is on the label dimension, and their beam has no greedy floor, so a narrow width there risks falling *below* the local baseline — but the direction of the finding agrees.

A label-dimension cap was considered as the next move and **rejected**: the joint arm sits at P=0.61 / R=0.15, so precision is four times recall, and a pruner targets the axis already being won. Compute is not binding either — W=64 ran in the same wall clock as W=1.

4.2 The data-scaling curve

Four base sizes, re-run from scratch on repaired data. Strict micro-F1 on the held-out blind test; the pool is 136,772 records at the largest point.

head	10K	40K	100K	137K
event_type	0.8138	0.7503	0.7979	**0.9841**
event_trigger	0.2259	0.3285	0.4032	**0.6953**
classification	0.0974	0.4838	0.5518	**0.6336**
entity	0.2779	0.4323	0.4795	**0.5158**
event	0.1475	0.2232	0.2643	**0.2688**
relation	0.0058	0.0175	0.0486	**0.2071**
event_argument	0.0130	0.0502	0.0815	0.0692
structure †	0.0294	0.0568	0.1102	**0.1119**

† The structure row is measured on the full test splits at each model's own swept record threshold; see §4.4. Every other row is at the shared decision threshold.

The 100K → 137K jump is recall unlocking, not new capability. Only 1.37× the data, but event_trigger gains +0.29, relation +0.16 and event_type +0.19. The precision/recall split gives it away: event_trigger at 100K was P=0.58 / R=0.31, and at 137K P=0.62 / R=0.79. The smaller models were not bad at the task

— they were *withholding predictions*. Read as smooth capability scaling, this curve supports the wrong conclusion about what more data buys.

`event_argument` strict *falls* 0.0815 → 0.0692 while its relaxed score rises 0.2758 → 0.5064: the same inversion. More arguments proposed, approximately right far more often, exact spans lagging. Calibration, not regression.

4.3 Warm start with 30% exact replay

The 137K base was warm-started on new-domain data (real news plus synthetic, 21,354 records) with 9,151 replay records drawn from **the base's own training pool**, then scored on the base's own blind test:

head	base 137K	warm start	delta
entity	0.5158	**0.6326**	**+0.1168**
event	0.2688	**0.3644**	**+0.0956**
event_trigger	0.6953	**0.7490**	**+0.0537**
event_argument	0.0692	0.0975	+0.0283
classification	0.6336	0.6394	+0.0058
structure	0.1119	0.1060	−0.0059
event_type	0.9841	0.9447	−0.0394
relation	0.2071	0.1245	−0.0826

With 30% replay the model gained on the original task while learning a new one. For context, three zero-replay fine-tunes of a comparable base on the same corpora lost 23%, 32% and 39% of general-domain entity F1, in monotonic order of training volume.

This arm is stronger evidence than those because we own both the base and its pool, so the replay is **exact** rather than a proxy drawn from a stand-in corpus.

Qualifications that must travel with the result: it is a different architecture from the zero-replay arms, so it is not a controlled contrast; part of the gain is genuinely new learning, since the new corpora supply real entity signal; and **protection is not uniform** — relation −0.083 and event_type −0.039 both declined, which is the open thread.

4.4 The structure head: what it does once it is asked

With the decode defect of §3.3 fixed, the same checkpoints were re-scored — nothing retrained — at the record head's own swept thresholds:

	10K	40K	100K	137K
structure strict F1	0.0294	0.0568	0.1102	**0.1119**
at record threshold	0.03	0.05	0.10	0.10

A monotone curve that flattens hard between 100K and 137K (+0.0017), where 100K is actually the more precise model (0.303 versus 0.237). This is the ordinary shape of a head learning from 7,754 supervised records. There was never an anomaly to explain — only a question the head was never asked.

Recall is the binding constraint throughout: even the best model recovers 7.3% of gold triples.

The warm start is the sharper result. Scoring both checkpoints on both test sets gives the full 2×2, which is the only readable form:

scored on	base 137K	warm start	delta
137K test set (856 records)	**0.1119**	0.1060	−0.0059
warm-start test set (452 records)	0.0755	**0.2179**	**+0.1424**

Read down the columns, never across the rows — the two test sets differ in difficulty and distribution. The warm start **held the old capability (−5% relative) and nearly tripled on the new distribution (+189%)**. The single-test-set reading, that it merely survived, could not see the gain.

The mechanism matters more than the number. The new corpora contributed **zero** record-head targets, because they emit structures without the metadata the training path requires (§3.3): the record head saw 519 replay records against the base's 7,754, a 93% *reduction*. The +0.1424 is therefore not learned record structure but **transfer from the span representations the record head reads its field fillers out of** — entity F1 moved 0.5158 → 0.6326 over the same run. Supplying real metadata for those corpora is a measurable next experiment rather than a speculative cleanup.

5. Limitations

- **The relation result is one model on one corpus.** Joint-versus-greedy is measured on Re-DocRED only. The event face of the same decode is not yet measured, so the thesis is supported on one of its two faces.
- **Single-run variance on these metrics is ±0.02 or worse.** A control re-run of a published recipe scored +0.023 above it from the re-run alone. The +0.052 joint-decode gain clears that floor; smaller deltas quoted anywhere in this paper do not, and are flagged where they appear.
- **The scaling curve is one seed per point.** Its *shape* is informative and its point-to-point differences at the 100K → 137K step are not.
- **`event_argument` at 137K sits below its 100K value** on strict, and the calibration reading of that inversion is an interpretation, not a measurement.
- **The warm start's protection is not uniform.** Relation and event_type both declined; no mechanism is established for why those two and not the others.
- **Structure supervision is not reaching the record head from most corpora** in the 137K configs above, so any arm claiming structure capability from them added none. **Resolved 2026-09-08 (`RESEARCH_PROGRAM.md` §3):** all 60,948 affected rows across 13 corpora and the synthetic validator were repaired to declare `record_metadata` (`mode` and `anchor`), gated by `audit_corpora.py`.

- **Events in the 10K–137K configs above were supervised through the mention-structure path (`event_records: false`), not the record head.** That pooled multi-instance same-type events into a single record per type (collapsing 64.2% of multi-instance gold events) and explained why `event_argument` strict micro-F1 stalled at `0.069–0.098` in §4.2–4.3 while relaxed micro-F1 reached `0.506`. **Resolved in Paper 3 (`PAPER_3_DOCUMENT_EVENTS_AND_COREF.md`):** enabling `event_records: true`, adding the gated bilinear + geometry junction layer (`record_link_mode: junction`), training coreferent trigger and argument clusters (`TriggerLink`, `EntityCorefLink`, `record_coreferent_ownership: true`), and fixing positive-gradient starvation in both `_list_field_bce` and `candidate_pair_loss` (`eb22g`) lifts 64-doc overfit `@0.5` argument strict micro-F1 to `0.4319` on CASIE and `0.8005` on CMNEE, and lifts full-mix Epoch 2 validation `event_argument` strict micro-F1 to `0.1639` (relaxed `0.3992`, fair `0.3632`).

6. Reproducibility

Configurations are in `tools/train/config/`: `joint-boundary-mmbert-{10k,40k,100k,137k}.yaml` for the curve, `warmstart-137k-realsynth-replay30.yaml` for the replay arm. Replay slices are built by `tools/train/build_137k_replay.py` (proportional across all 13 pool corpora, seed 42). Record-head thresholds are swept by `tools/train/sweep_record_thresholds.py`, which no earlier tool did — the trainer's threshold sweep moves the shared decision threshold and never touches the record cutoffs. Split integrity is gated by `tools/data/check_leakage.py --config`, wired into `tools/train/train.py` so it runs before every training run.

Models are published privately as `whr778/gliner2-joint-boundary-mmbert-{10k,40k,100k,137k}-clean` and `whr778/gliner2-warmstart-137k-realsynth-replay30`. Each model card reports the structure metric at its own swept record threshold alongside what the 0.5 default would have scored.

The build record — decision log, wiring map, increment status, data survey, cost model and the deferred Phase B plan — is in `JOINT_IE_DESIGN_RECORD.md`.

7. References

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